WEBSITE CONVERSION FUNNEL ANALYSIS

Report

Python + SQL Analysis of 90,400 Users

90,400	452	0.50%	92.50%
Users analysed	Completed purchases	Overall conversion	Final-stage abandonment

Executive Summary

This analysis reviewed the purchase journey of 90,400 users to understand where potential customers were being lost and which areas should be improved first.

Finding	Result	Why it matters
Overall conversion	0.50%	Only 452 of 90,400 users completed a purchase.
Largest user loss	39,170 users	Most users were lost between Search and Payment.
Highest abandonment	92.50%	Payment to Confirmation was the weakest transition proportionally.
Largest segment opportunity	Desktop	Desktop held 66.6% of users but converted at only 0.25%.

The main opportunity is not simply to attract more traffic. It is to convert more of the users already entering the funnel. The business should first investigate payment completion, then improve the Desktop purchase journey, and finally strengthen the move from Search to Payment.

Business Objective

The business needed to know where conversion leakage was highest, which user groups were most affected, and where product, marketing, and checkout improvements would have the greatest practical impact.

Data and Approach

The analysis used five related tables covering users and the four funnel stages: Home, Search, Payment, and Payment Confirmation. Python was used for detailed analysis and visualisation, while SQL was used to validate the data and build a reusable user-level model with one record per user.

- 90,400 users were reviewed across five relational tables.
- Each user was assigned stage indicators and a clear drop-off outcome.
- Performance was compared overall, by device, and across monthly user groups.
- All SQL outputs reconciled to the original source counts.

1. Funnel Performance

The funnel narrowed sharply at every stage. Of the 90,400 users who entered the journey, only 452 completed a purchase.

Transition	Users reached	Users progressed	Conversion	Abandonment
Home → Search	90,400	45,200	50.00%	50.00%
Search → Payment	45,200	6,030	13.34%	86.66%
Payment → Confirmation	6,030	452	7.50%	92.50%

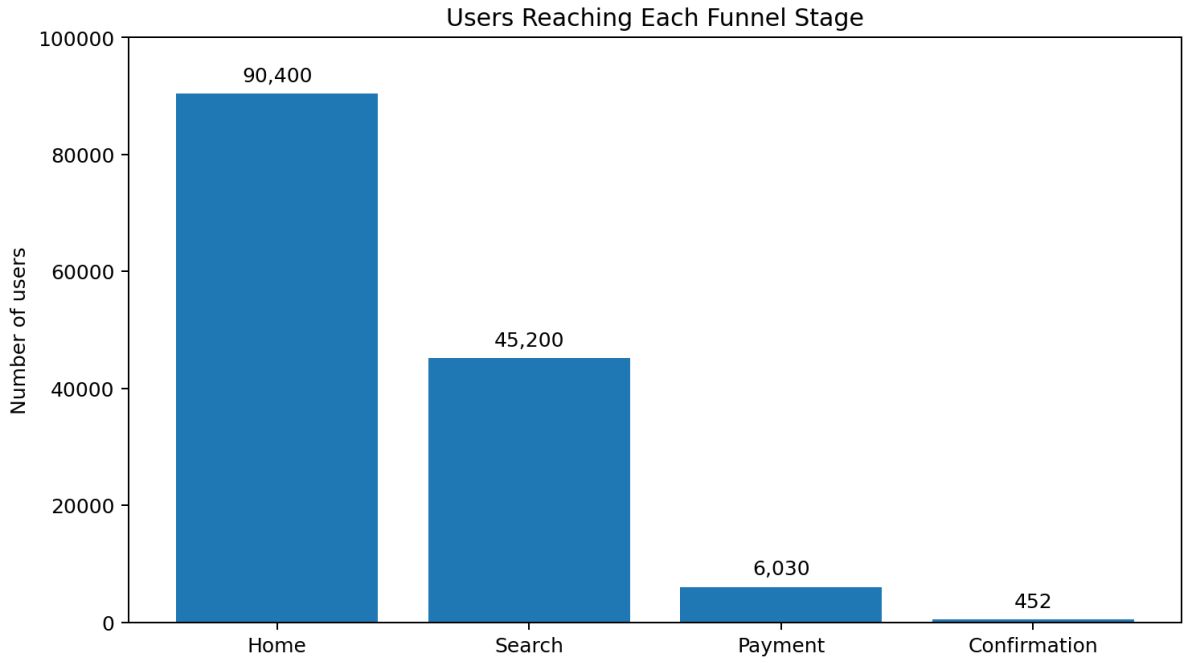


Figure 1. Users reaching each stage of the purchase funnel

What the numbers mean

Search → Payment caused the largest loss in volume: 39,170 users left before reaching the payment page. This is the biggest area of leakage by number of users.

Payment → Confirmation had the highest abandonment rate: 92.50% of users who reached Payment did not complete the purchase. These users were already close to buying, so this is the most urgent stage to investigate.

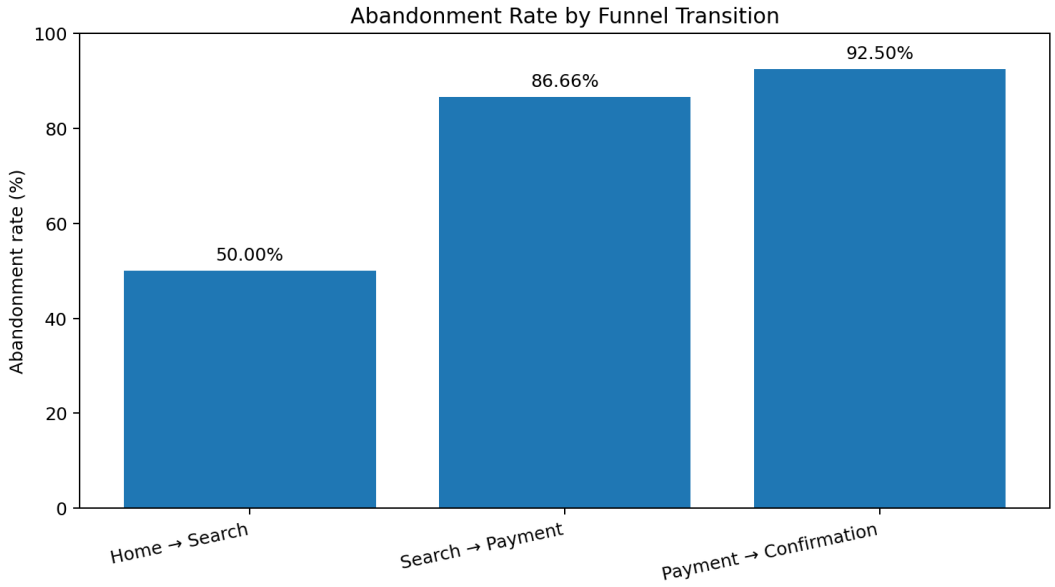


Figure 2. Abandonment rate by funnel transition

2. Device Performance

Desktop was the largest user segment, but it performed much worse than Mobile at the end of the funnel.

Device	Users	Share of users	Overall conversion
Desktop	60,200	66.6%	0.25%
Mobile	30,200	33.4%	1.00%

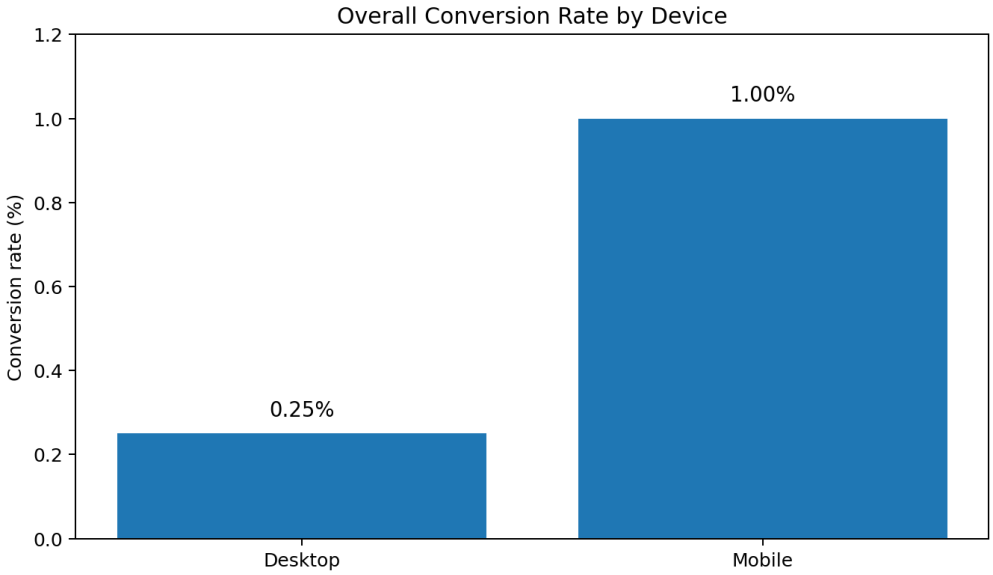


Figure 3. Overall conversion rate by device

Both devices achieved the same 50.00% Home → Search conversion, suggesting similar interest at the start of the journey. Desktop converted at only 0.25%, compared with 1.00% on Mobile.

Because Desktop represents 60,200 users, even a small improvement could affect a large existing audience without requiring additional traffic.

3. Other Segment and Time Insights

Gender

Male and Female users showed only small differences in conversion. Gender is therefore not a major optimisation priority compared with funnel stage and device performance.

Monthly user groups

Users associated with March and April showed weaker funnel performance than users associated with January and February. The available data does not explain why this happened, so these groups should be investigated further rather than treated as proof of a specific time-based decline.

The stage tables do not include event timestamps. Monthly comparisons are based on the date in the main users table, not the exact time at which each funnel action occurred.

4. Recommended Actions

Priority 1: Investigate payment completion

Payment → Confirmation recorded 92.50% abandonment.

- Review payment failure logs, OTP issues, bank declines, and gateway errors.
- Simplify checkout steps and collect feedback from users who could not complete payment.

Priority 2: Improve the Desktop journey

Desktop holds 66.6% of users but converts at only 0.25%.

- Review the full Desktop funnel and conduct usability testing.
- Simplify forms and navigation, then validate improvements through A/B testing.

Priority 3: Strengthen Search → Payment

This transition lost 39,170 users, the largest absolute drop-off.

- Improve search relevance, product information, trust signals, and offer visibility.
- Test personalised recommendations and clearer pricing information.

How success should be measured

- Payment → Confirmation conversion rate
- Number of completed purchases
- Desktop overall conversion rate
- Search → Payment conversion rate
- Payment failure and checkout abandonment reasons

5. Limitations

- The data shows where users leave the funnel, but not the exact reason for leaving.
- Payment failure logs, website behaviour, pricing, campaign source, and customer feedback were not available.
- The stage tables did not contain event timestamps, so the monthly analysis is based on the date in the users table.
- The dataset did not include revenue information, so the financial impact of conversion improvements could not be quantified.

Conclusion

The analysis found that the largest growth opportunity lies in improving conversion among existing users. Payment completion should be investigated first because it has the highest abandonment rate, while the Desktop journey should be the main segment priority because it combines the largest audience with the weakest conversion performance.

Improving these areas could increase completed purchases without proportionally increasing customer-acquisition spending. Any changes should be tested and measured before full implementation.